

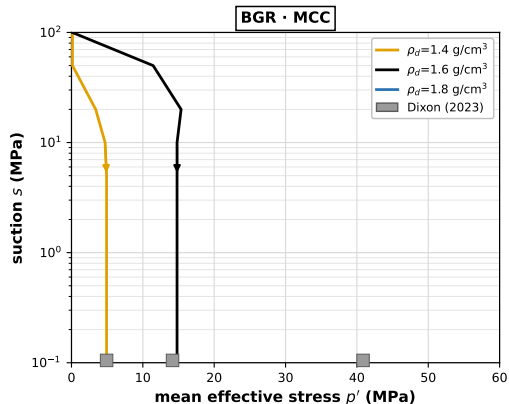
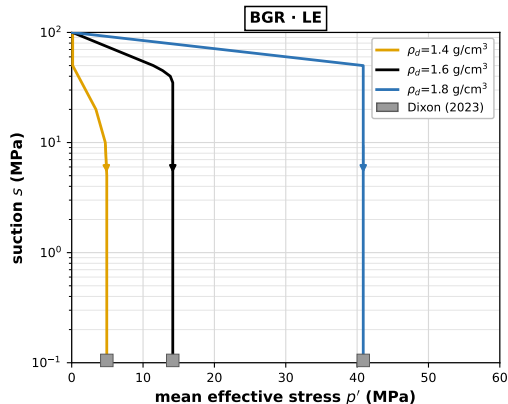
BGR MS33 — FASTEX recalibration: LE vs MCC figure review

constant $k_{rel} = 0.1$ (temporary) $\cdot \alpha_{exch} = 3 \times 10^{-13} \cdot K \rightarrow$ Dixon EMDD $\equiv \rho_d$
(2026-06-02)

Calibration summary

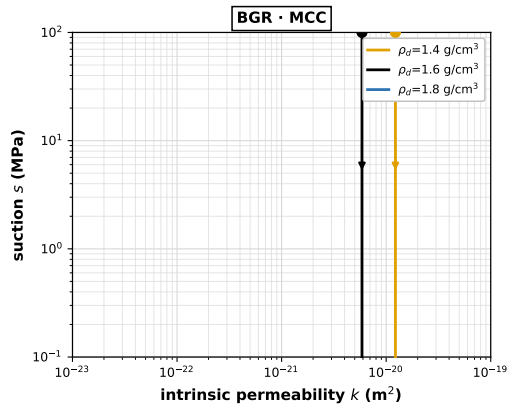
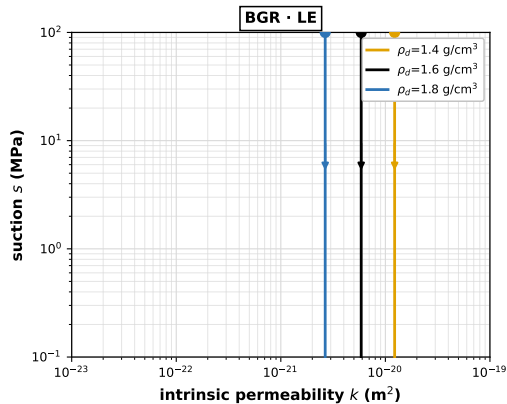
- ▶ $k_{rel} = \mathbf{0.1 \text{ (constant)}}$ — fixes the dry-macro transport throttle (was S_e^3 floored at 0.01). *Temporary; reverts to orthotropic power law.*
- ▶ $\alpha_{exch} = 3 \times 10^{-13}$ — the transient: Model I $t_{95} = 18/12/10$ d (<20 d); III flattens ~ 50 d.
- ▶ K (*vdw_augmentation_prefactor*) — to Dixon EMDD $\equiv \rho_d$: dd1.4=34368, dd1.6=83377, dd1.8=224610 J/kg ($\rightarrow 4.92/14.16/40.86$ MPa). III/IV/VII inherit dd1.6 ($\times 0.97729$).
- ▶ Each figure is tagged **BGR · LE** or **BGR · MCC**.

Model I — mean stress vs suction path (p' 0–60 lin, s log 0.1–100)



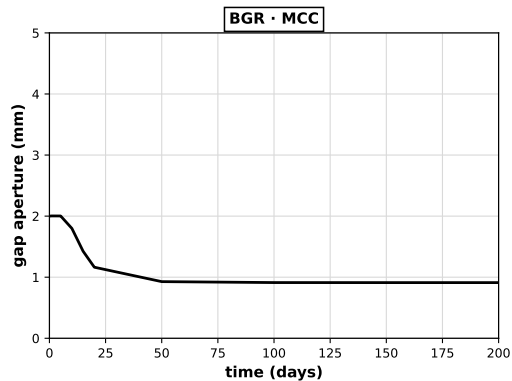
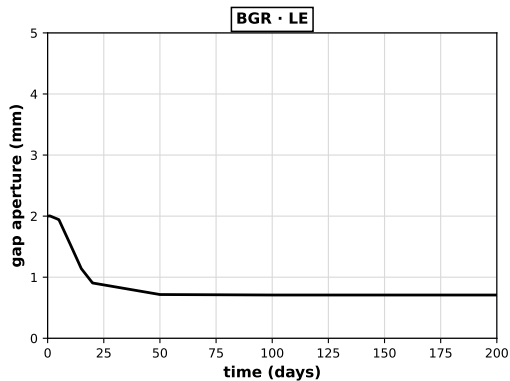
LE: all three dd on Dixon (dd1.8 now *completes* 200d via the power-law k_{rel} exception — a single element needs no transport floor). MCC: dd1.4 on Dixon, dd1.6 *yields* (~ 15 MPa, kinked), dd1.8 absent (crash@0 $\rightarrow$ bare Dixon square).

Model I — intrinsic permeability vs suction (k log 1e-23–1e-19, s log)



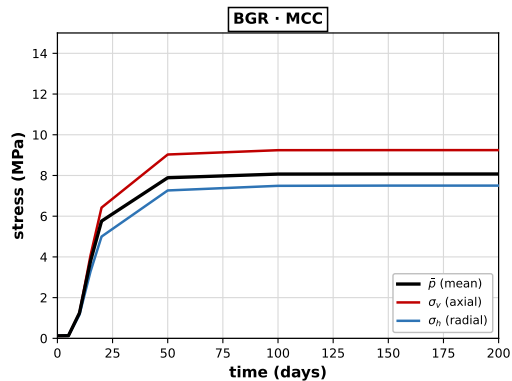
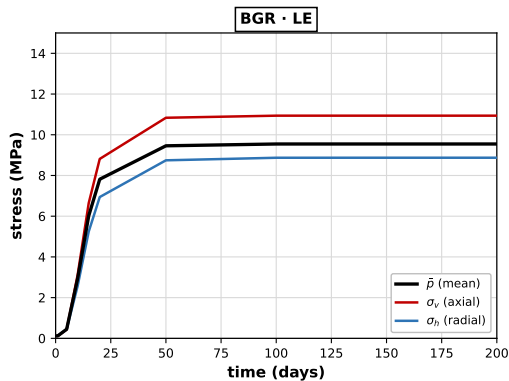
k near-constant (Kozeny–Carman; porosity fixed under constant-volume swelling) — vertical lines, one per dry density.

Model III — technological-gap closure (aperture mm vs time 0–200 d)



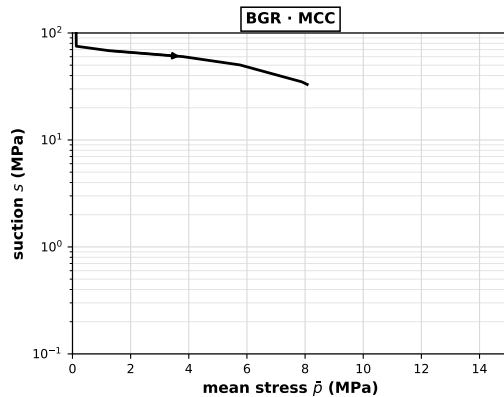
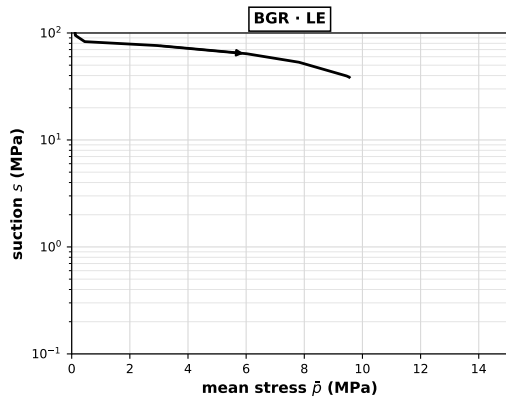
Gap closes 2.0 → 0.71 mm (LE) / 0.91 mm (MCC).

Model III — stress evolution at centre (mean/axial/radial, 0–15 MPa vs time)

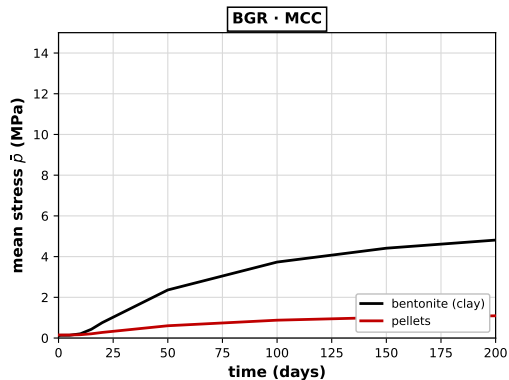
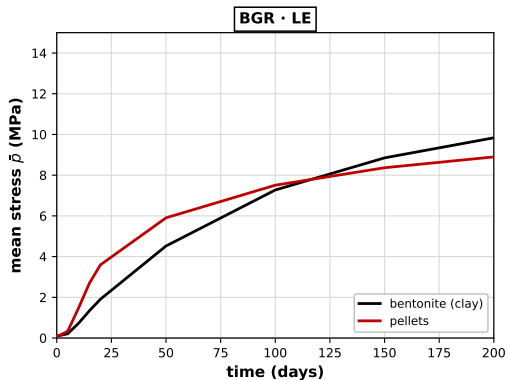


Now flattens (the k_{rel} fix) — was creeping past 200 d before. MCC III converges where it crashed at the tension apex previously.

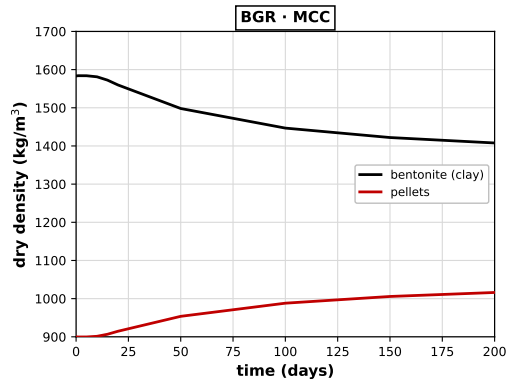
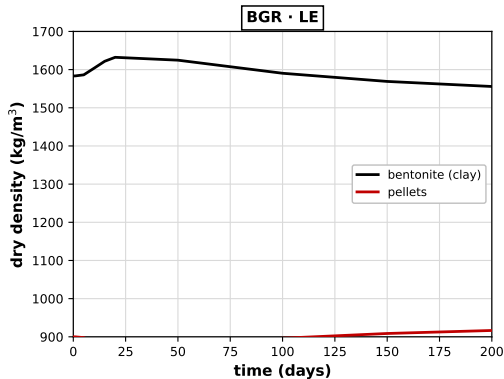
Model III — mean stress vs suction path ($\bar{p}$ 0–15, s log)



Model IV — mean stress per zone (clay / pellet, 0–15 MPa vs time)

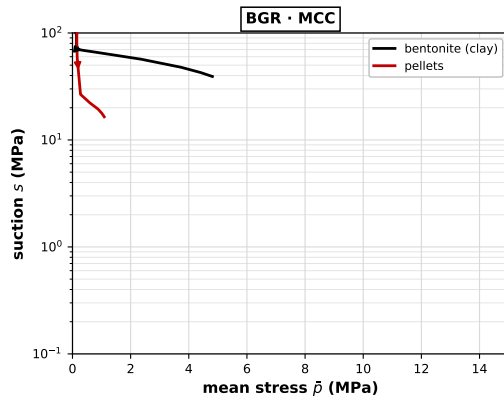
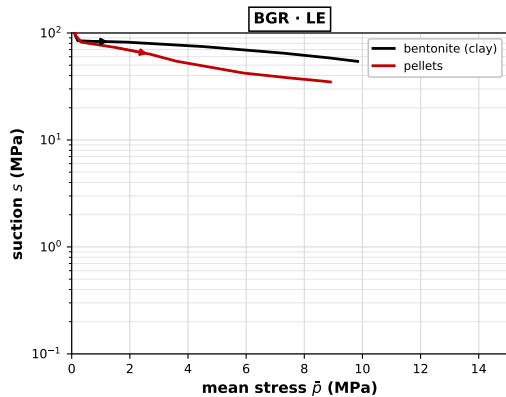


Model IV — dry-density homogenisation (clay / pellet, 900–1700 kg/m³ vs time)

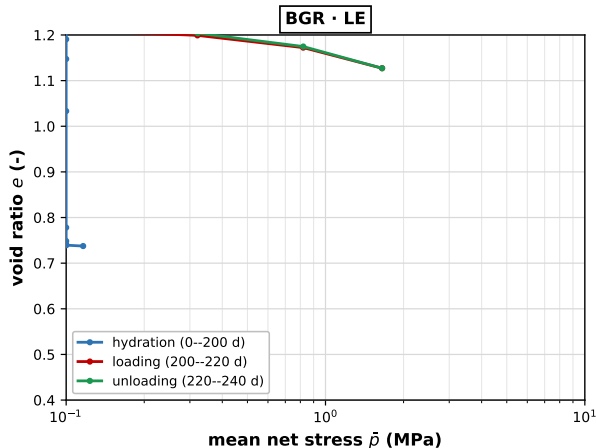


MCC homogenises (clay 1585→1408, pellet 900→1016 — converging, BExM-style); **LE does not** (clay/pellet barely move). The double structure carries it.

Model IV — mean stress vs suction path per zone ($\bar{p}$ 0–15, s log)

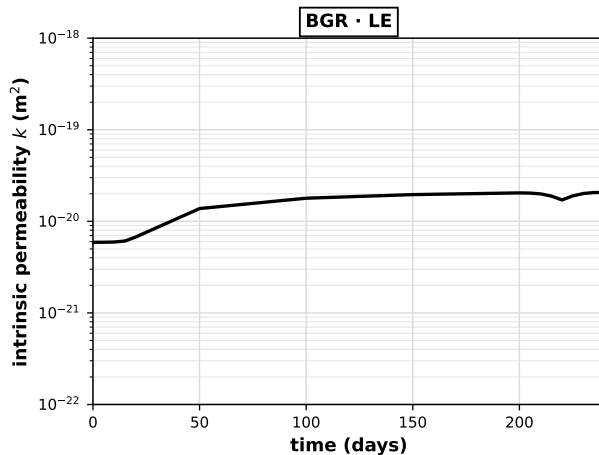


Model VII — void ratio vs mean net stress (e 0.4–1.2, $\bar{p}$ log 0.1–10)



LE: free-swell $\rightarrow$ load $\rightarrow$ unload loop (phase-coloured). **MCC VII: genuinely broken** — upstream MFront cam-clay return-map at the tension apex ($p_{net} \rightarrow 0$ with residual q , status -1), *not* k_{rel} . LE e peaks at 1.22 (just over the spec axis top).

Model VII — intrinsic permeability evolution (k log, vs time)



LE: k rises with the free-swelling porosity increase. **MCC VII: no converged run.**

	dd1.4	dd1.6	dd1.8	III	IV	VII
LE	Dixon −0.1%	Dixon +0.0%	Dixon (PL)*	flat	done	loop
MCC	Dixon −0.1%	yields	crash@0	flat	homog.	crash@0

* LE dd1.8 *resolved*: power-law k_{rel} (min 0.1) on the single element completes 200 d at $K=224610 \rightarrow 40.86$ MPa (Dixon), no K re-fit. MCC dd1.8 / VII confirmed *upstream MFront* cam-clay return-map (high K / tension apex, status −1) — *not* k_{rel} , genuinely broken. Others use the temporary constant $k_{rel}=0.1$.